SCS 1311 – Internet and Web Technologies

HTML

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What Are Attributes?

- Attributes give extra information about an element

```
attribute="value"
```

Example -

```
<img src="photo.jpg" alt="A photo">
```

Common Attributes

Attribute	Used For	Example
src	image/video source	`<img src="cat.jpg" width="300" height="200" alt="A cute cat">`
width	Width of element	
height	Height of element	
alt	image description	

Common Attributes

Attribute	Used For	Example
href	link target	`<a href="https://example.com">Go to Example</a>`
type	type of input	`<input type="password">`
value	input value	`<input type="text" value="Hello">`

Attribute Types

1. **Global Attributes**

- Can be applied to any HTML element
- Provide universal functionality like identification, styling, or accessibility
  - eg- id, class, title

2. **Local (Element-Specific) Attributes**

- Attributes that only apply to certain elements
- Control behavior unique to that element type
  - eg- src, alt, width

Attribute Types

3. Boolean Attributes

- Attributes that are either present or absent and no value needed.
- Switch a feature on/off
  - eg- required, autofocus, disabled

Attribute Types

```
<body>
  <!-- Global attributes: id, class, title -->
  <p id="intro" class="highlight" title="Extra info">Sample text</p>

  <!-- Local attributes: src, alt, width, height (specific to <img>) -->
  <img src="example.jpg" alt="Tree" width="300" height="200">

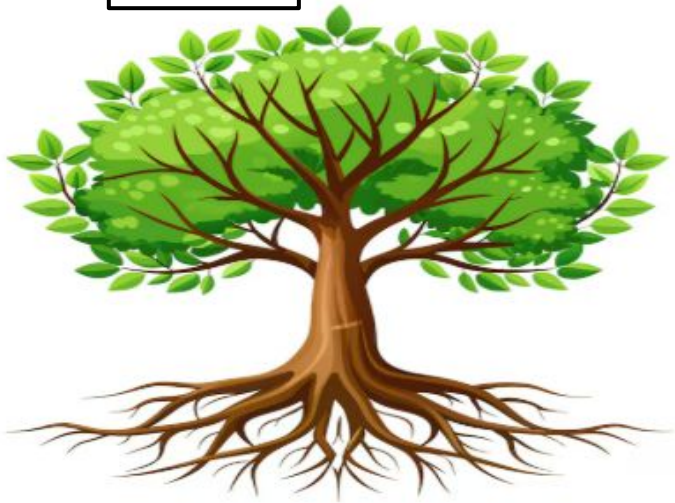
  <!-- Boolean attributes: disabled, required, autofocus -->
  <form>
    <input type="email" placeholder="Enter email" required>
    <input type="text" value="Disabled field" disabled>
    <input type="text" placeholder="Focused field" autofocus>
    <button type="submit">Submit</button>
  </form>

</body>
```

Attribute Types

Sample text

Extra Info



Enter email

Please fill out this field.

Disabled field

Focused field

Submit

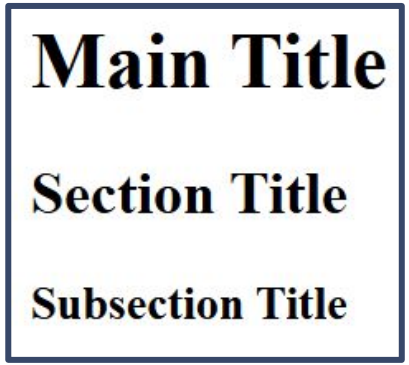
Text & Headings

- HTML provides 6 levels of headings
- <h1> is the most important heading
- <h6> is the least important
- Headings **define content hierarchy, not size**
- Search engines use headings to understand structure

```
<h1>Main Title</h1>

<h2>Section Title</h2>

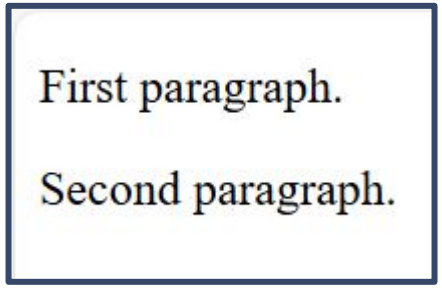
<h3>Subsection Title</h3>
```



Paragraph

- <p> defines a block of text
- Browsers add automatic spacing
- Only for meaningful paragraphs, not layout

```
<p>First paragraph.</p>
<p>Second paragraph.</p>
```



Line Breaks and Horizontal Rules

**
 - Line Break**

- Inserts a single line break
- Use only for small line breaks (poetry, addresses)

```
<p>Line one<br>Line two</p>
```

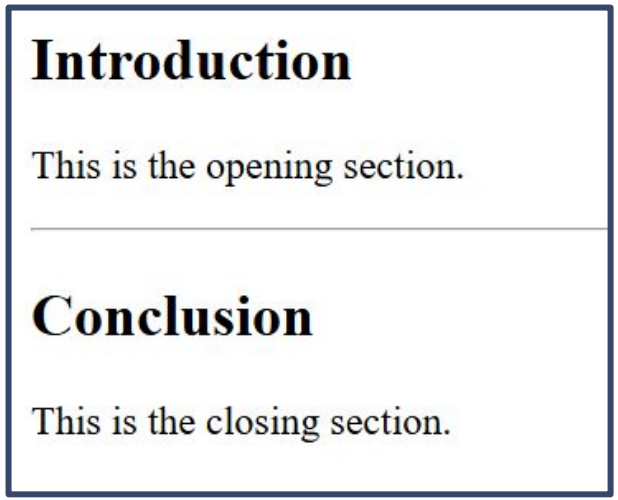
Line one
Line two

Line Breaks and Horizontal Rules

<hr> - Thematic Break

- Creates a horizontal rule (line) that represents a thematic division in content

```
<h2>Introduction</h2>
<p>This is the opening section.</p>
<hr>
<h2>Conclusion</h2>
<p>This is the closing section.</p>
```



Inline Text Formatting (Semantic Tags)

- Semantic inline elements:
  - `<strong>` – strong importance
  - `<em>` – emphasis
  - `<u>` – underline
  - `<mark>` – highlight
  - `<small>` – side text / fine print
  - `<sub>` – subscript
  - `<sup>` – superscript

Inline Text Formatting (Semantic Tags)

```
<p>This is <strong>important</strong>
information.</p>
<p><u>Water formula</u>: H<sub>2</sub>O</p>
<p>4<sup>2</sup> = 16</p>
<p>This is <mark>highlighted</mark>
text.</p>
<p><small>Note: This is additional fine
print.</small></p>
```

Inline Text Formatting (Semantic Tags)

This is **important** information.

Water formula: H_2O

$4^2 = 16$

This is highlighted text.

Note: This is additional fine print.

Non-Semantic Inline Elements

- — a generic inline container
- Used when,
  - You need to select part of text
  - The text has no special meaning

```
<p>This is <span class="highlight">special</span>
text.</p>
```

Block vs Inline (Text Context)

- Block-level elements,
  - Start on a new line
  - Take full width by default
  - Structure the page layout
- Inline elements,
  - Do not start on a new line
  - Take only the space needed
  - Are used inside text content

HTML Lists

Why Lists are important?

- Can present information in a structured and easy-to-read format
- Improve readability
- 3 types of lists
  - Unordered
  - Ordered
  - Description

Unordered Lists

- Used to mark up lists of items for which the **order of the items doesn't matter**
- The unordered list items are marked with bullets, also known as **bulleted lists**
- An unordered list starts with the **<ul>** tag, and each list item starts with the **<li>** tag

Unordered Lists <ul> Cont.

```
<ul>
    <li>List item 1</li>
    <li>List item 2</li>
    <li>List item 3</li>
</ul>
```

- List item 1
- List item 2
- List item 3

Ordered Lists

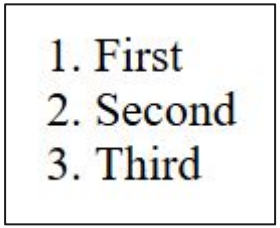
- Used when the items need to follow a specific sequence.
- An ordered list starts with the **<ol>** tag, and each list item starts with the **<li>** tag.
- All list items are marked with numbers by default
- Attributes,
  - type -> Numbering style
  - start -> Starting number
  - reversed -> Reverse numbering (boolean)

Ordered Lists <ol> Cont.

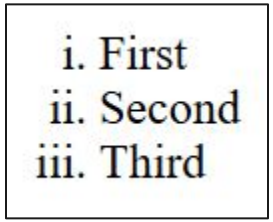
- type - controls the numbering style
  - 1 ⟶ 1, 2, 3 (default)
  - a ⟶ a, b, c
  - A ⟶ A, B, C
  - i ⟶ i, ii, iii
  - I ⟶ I, II, III

Ordered Lists <ol> Cont.

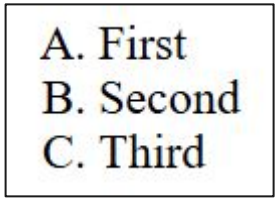
```
<ol>
    <li>First</li>
    <li>Second</li>
    <li>Third</li>
</ol>
```




```
<ol type="i">
    <li>First</li>
    <li>Second</li>
    <li>Third</li>
</ol>
```




```
<ol type="A">
    <li>First</li>
    <li>Second</li>
    <li>Third</li>
</ol>
```



Ordered Lists <ol> Cont.

- start- set the starting number
- Any integer number is allowed
  - Positive integers (most common)
  - Zero
  - Negative integers (rarely used)

```
<ol start="3">
  <li>Third</li>
  <li>Fourth</li>
</ol>
```

3. Third
4. Fourth

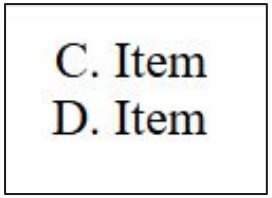
Ordered Lists <ol> Cont.

How **start** works with **type?**

Type	Meaning of start=”4”
1	starts at 4
A	starts at D
a	starts at d
I	starts at IV
i	starts at iv

Ordered Lists <ol> Cont.

```
<ol type="A" start="3">
   <li>Item</li>
   <li>Item</li>
</ol>
```



Ordered Lists <ol> Cont.

- reversed - displays the list in descending order.

```
<ol reversed>
  <li>One</li>
  <li>Two</li>
</ol>
```

2. One
1. Two

```
<ol start="5" reversed>
  <li>Item</li>
  <li>Item</li>
</ol>
```

5. Item
4. Item

```
<ol type="A" reversed>
  <li>Item</li>
  <li>Item</li>
  <li>Item</li>
</ol>
```

C. Item
B. Item
A. Item

```
<ol type="i" start="4" reversed>
  <li>Item</li>
  <li>Item</li>
  <li>Item</li>
</ol>
```

iv. Item
iii. Item
ii. Item

Nesting Lists

- A nested list is a **list inside a list item**
- Used to represent hierarchy
- Nesting happens inside
- The nested list can be:
  - the same type
  - a different type

Nesting Lists Cont.

```
<ol>
  <li>Step One
    <ol>
      <li>Sub step 1</li>
      <li>Sub step 2</li>
    </ol>
  </li>
  <li>Step Two</li>
</ol>
```

1. Step One
    1. Sub step 1
    2. Sub step 2
2. Step Two

Nesting Lists Cont.

```
<ul>
  <li>Step One
    <ul>
      <li>Sub step 1</li>
      <li>Sub step 2</li>
    </ul>
  </li>
  <li>Step two</li>
</ul>
```

- Step One
  - Sub step 1
  - Sub step 2
- Step two

Nesting Lists Cont.

```
<ol>
  <li>Step One
    <ul>
      <li>Sub step one</li>
      <li>Sub step one</li>
    </ul>
  </li>
  <li>Step Two</li>
</ol>
```

1. Step One
    - Sub step one
    - Sub step one
2. Step Two

Description Lists <dl>

- Purpose of description lists is to mark up a set of items and their associated descriptions
  - terms and definitions
  - questions and answers

- <dl> → container for the whole list
- <dt> → definition term (the keyword)
- <dd> → description (explanation of the term)

Description Lists <dl>

```
<dl>
    <dt>Item 1</dt>
    <dd>Description of Item 1 </dd>
    <dt>Item 2</dt>
    <dd>Description of Item 2</dd>
</dl>
```

Item 1
    Description of Item 1
Item 2
    Description of Item 2

Tables

- Tables are used to display tabular data
- Data is arranged in rows and columns
- An HTML table consists of:
  - `<table>` → table container
  - `<tr>` → table row
  - `<td>` → table data (cell)

Table Rows <tr> and Data Cells <td>

- <tr> defines one row
- Contains one or more cells
- Each row should have the same number of cells
- <td> represents a single data cell
- Can contain,
  - Text
  - Images
  - Links
  - Even other HTML elements

Tables Cont.

```
<table>
  <tr>
    <td>Name</td>
    <td>Age</td>
  </tr>
  <tr>
    <td>Nimal</td>
    <td>20</td>
  </tr>
</table>
```

Name Age
Nimal 20

- Table data appears
- **No visible lines**

Tables Cont.

- HTML tables have no borders by default
- HTML defines structure, not appearance
- Borders are part of styling
  - Old Way: **`border`** Attribute (Legacy)
  - Modern Way: Using CSS

Tables Cont.

```
<table border=”1”>
  <tr>
    <td>Name</td>
    <td>Age</td>
  </tr>
  <tr>
    <td>Nimal</td>
    <td>20</td>
  </tr>
</table>
```

Name	Age
Nimal	20

```
<table style="border: 1px solid black;">
  <tr>
    <td>Name</td>
    <td>Age</td>
  </tr>
  <tr>
    <td>Nimal</td>
    <td>20</td>
  </tr>
</table>
```

Name	Age
Nimal	20

Tables Cont.

```
<table style="border: 1px solid black;">
  <tr>
    <td style="border: 1px solid black;">Name</td>
    <td style="border: 1px solid black;">Age</td>
  </tr>
  <tr>
    <td style="border: 1px solid black;">Nimal</td>
    <td style="border: 1px solid black;">20</td>
  </tr>
</table>
```

Name	Age
Nimal	20

Table Headers

- <th> defines **header cells**
- Used for headings of rows or columns
- Displayed **bold and centered** by default
- Improves accessibility

```
<tr>
  <th>Name</th>
  <th>Marks</th>
</tr>
```

Column Headers vs Row Headers

```
<tr>
  <th>Name</th>
  <th>Marks</th>
</tr>
<tr>
  <td>Nimal</td>
  <td>18</td>
</tr>
<tr>
  <td>Kamal</td>
  <td>20</td>
</tr>
```

Name	Marks
Nimal	18
Kamal	20

```
<tr>
  <th>Name</th>
  <td>Nimal</td>
  <td>Kamal</td>
</tr>
<tr>
  <th>Marks</th>
  <td>18</td>
  <td>20</td>
</tr>
```

Name	Nimal	Kamal
Marks	18	20

Thank You!